The following theorems by E. Lucas in 1878 and E. Kummer in 1852 are very useful in number theory. Let n be a positive integer, and let p be a prime. Let $(\overline{n_m n_{m-1} \dots n_0})_p$ denote the **base p representation of n**; that is,

$$n = (\overline{n_m n_{m-1} \dots n_0})_p = n_0 + n_1 p + \cdots + n_m p^m,$$

where $0 \leq n_0, n_1, \dots, n_m \leq p-1$ and $n_m \neq 0$.

Theorem 3.3. *[Lucas] Let p be a prime, and let n be a positive integer with $n = (\overline{n_m n_{m-1} \dots n_0})_p$. Let i be a positive integer less than n. Also, write $i = i_0 + i_1 p + \cdots + i_m p^m$, where $0 \leq i_0, i_1, \dots, i_m \leq p-1$. Then*

$$\binom{n}{i} \equiv \prod_{j=0}^{m} \binom{n_j}{i_j} \pmod{p}. \qquad (**)$$

Here $\binom{0}{0} = 1$ and $\binom{n_j}{i_j} = 0$ if $n_j < i_j$ by the convention we set earlier.

To prove this theorem, we need some new terminology. Let p be a prime, and let $f(x)$ and $g(x)$ be two polynomials with integer coefficients. We say that $f(x)$ is **congruent** to $g(x)$ modulo p, and write $f(x) \equiv g(x) \pmod{p}$ if all of the coefficients of $f(x) - g(x)$ are divisible by p. (Note that the congruence of polynomials is different from the congruence of the values of polynomials. For example, $x(x+1) \not\equiv 0 \pmod{2}$ even though $x(x+1)$ is divisible by 2 for all integers x.) The following properties can easily be verified:

(a) $f(x) \equiv f(x) \pmod{p}$;

(b) if $f(x) \equiv g(x) \pmod{p}$, then $g(x) \equiv f(x) \pmod{p}$;

(c) if $f(x) \equiv g(x) \pmod{p}$ and $g(x) \equiv h(x) \pmod{p}$, then $f(x) \equiv h(x) \pmod{p}$;

(d) if $f(x) \equiv g(x) \pmod{p}$ and $f_1(x) \equiv g_1(x) \pmod{p}$, then $f(x) \pm f_1(x) \equiv g(x) \pm g_1(x) \pmod{p}$ and $f(x)f_1(x) \equiv g(x)g_1(x) \pmod{p}$.

In other words, the congruence relation between polynomials is reflexive, symmetric, and transitive (so it is an equivalence relation), and is preserved under addition, subtraction, and multiplication.

Proof: By Theorem 3.2 (k), the binomial coefficients $\binom{p}{k}$, where $1 \leq k \leq p-1$, is divisible by p. Thus, $(1+x)^p \equiv 1 + x^p \pmod{p}$ and $(1+x)^{p^2} = [(1+x)^p]^p \equiv [1+x^p]^p \equiv 1 + x^{p^2} \pmod{p}$, and so on; so that for any positive integer r, $(1+x)^{p^r} \equiv 1 + x^{p^r} \pmod{p}$ by induction. We consider $(1+x)^n$.

We have

$$\begin{aligned}(1+x)^n &= (1+x)^{n_0+n_1p+\cdots+n_mp^m}\\ &= (1+x)^{n_0}[(1+x)^p]^{n_1}\cdots[(1+x)^{p^m}]^{n_m}\\ &\equiv (1+x)^{n_0}(1+x^p)^{n_1}\cdots(1+x^{p^m})^{n_m} \pmod p.\end{aligned}$$

The coefficient of x^i in the expansion of $(1+x)^n$ is $\binom{n}{i}$. On the other hand, because $i = i_0 + i_1p + \cdots + i_mp^m$, the coefficient of x^i is the coefficient of $x^{i_0}(x^p)^{i_1}\cdots(x^{p^m})^{i_m}$, which is equal to $\binom{n_0}{i_0}\binom{n_1}{i_1}\cdots\binom{n_m}{i_m}$. Hence

$$\binom{n}{i} \equiv \binom{n_0}{i_0}\binom{n_1}{i_1}\cdots\binom{n_m}{i_m} \pmod p,$$

as desired. ■

The proof of this theorem is yet another good application of generating functions, which will be discussed extensively in chapter 8.

Theorem 3.4. *[Kummer] Let n and i be positive integers with $i \le n$, and let p be a prime. Then p^t divides $\binom{n}{i}$ if and only if t is less than or equal to the number of carries in the addition $(n-i)+i$ in base p.*

For a positive integer m and a prime p, we write $p^{t_m}\|m$ if p^{t_m} divides m and p^{t_m+1} does not; that is, t_m is the greatest integer such that p^{t_m} divides m. Theorem 3.4 is based on the following two lemmas.

Lemma 3.5. *Let n be a positive integer, and let p be a prime. If $p^t\|n!$, then*

$$t = \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \left\lfloor \frac{n}{p^3} \right\rfloor + \cdots.$$

Proof: First we note that this sum is a finite sum, because for large m, $n < p^m$ and $\left\lfloor \frac{n}{p^m} \right\rfloor = 0$. Let m be the smallest integer such that $n < p^m$. It suffices to show that

$$t = \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \cdots + \left\lfloor \frac{n}{p^m} \right\rfloor.$$

For each positive integer i, define t_i such that $p^{t_i}\|i$. Because p is prime, we have $p^{t_1+t_2+\cdots+t_n}\|n!$, or $t = t_{n!} = t_1 + t_2 + \cdots + t_n$. On the other hand, $\left\lfloor \frac{n}{p^k} \right\rfloor$ counts all multiples of p^k that are less than or equal to n exactly once. Thus the number $i = p^{t_i} \cdot a$ (with a and p relatively prime) is counted t_i times in the sum

$$\left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \cdots + \left\lfloor \frac{n}{p^m} \right\rfloor,$$

namely, in the terms $\left\lfloor \frac{n}{p} \right\rfloor, \left\lfloor \frac{n}{p^2} \right\rfloor, \ldots, \left\lfloor \frac{n}{p^i} \right\rfloor$. Therefore, for each $1 \le i \le n$, the number i contributes t_i in both

$$\left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \cdots + \left\lfloor \frac{n}{p^m} \right\rfloor$$

and

$$t_1 + t_2 + \cdots + t_n.$$

Hence

$$t = t_1 + t_2 + \cdots + t_n = \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \cdots + \left\lfloor \frac{n}{p^m} \right\rfloor.$$

If you are still not convinced, consider the matrix $\mathbf{M} = (x_{i,j})$ with m rows and n columns, where m is the smallest integer such that $p^m > n$. We define

$$x_{i,j} = \begin{cases} 1 & \text{if } p^i \text{ divides } j, \\ 0 & \text{otherwise.} \end{cases}$$

Then the number of 1's in the j^{th} column of the matrix $\mathbf{M}$ is t_j, implying that the column sums of $\mathbf{M}$ are $t_1, t_2, \ldots, t_n$. Hence the sum of all of the entries in $\mathbf{M}$ is t. On the other hand, the 1's in the i^{th} row denote the numbers that are multiples of p^i. Consequently, the sum of the entries in the i^{th} row is $\left\lfloor \frac{n}{p^i} \right\rfloor$. Thus the sum of the entries in $\mathbf{M}$ is also $\sum_{i=1}^{m} \left\lfloor \frac{n}{p^i} \right\rfloor$. It follows that

$$t = \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \cdots + \left\lfloor \frac{n}{p^m} \right\rfloor,$$

as desired. ■

Lemma 3.6. *Let p be a prime, and let n be a positive integer with $n = (\overline{n_m n_{m-1} \ldots n_0})_p$. If $p^t \| n!$, then*

$$t = \frac{n - (n_m + n_{m-1} + \cdots + n_0)}{p-1}.$$